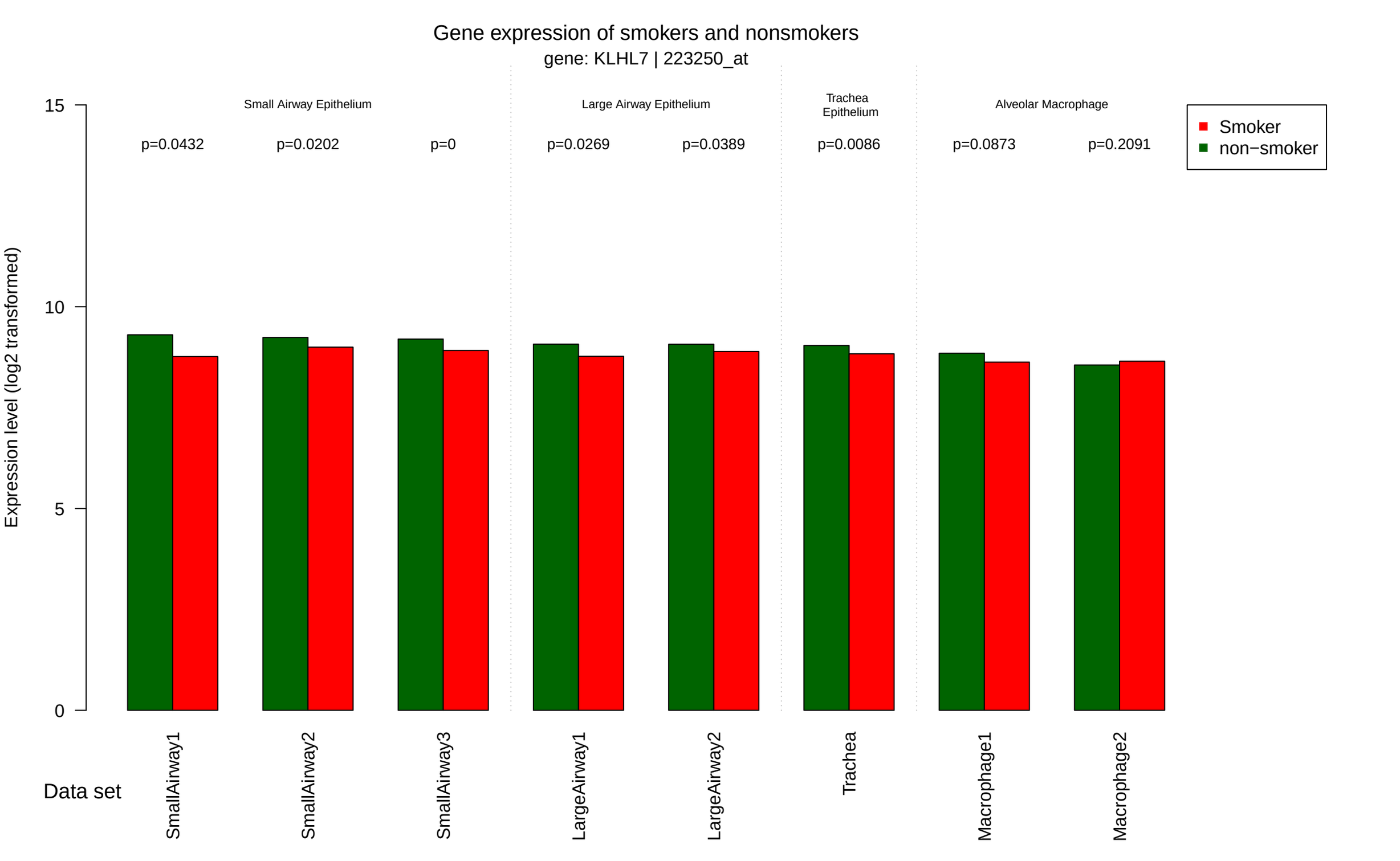
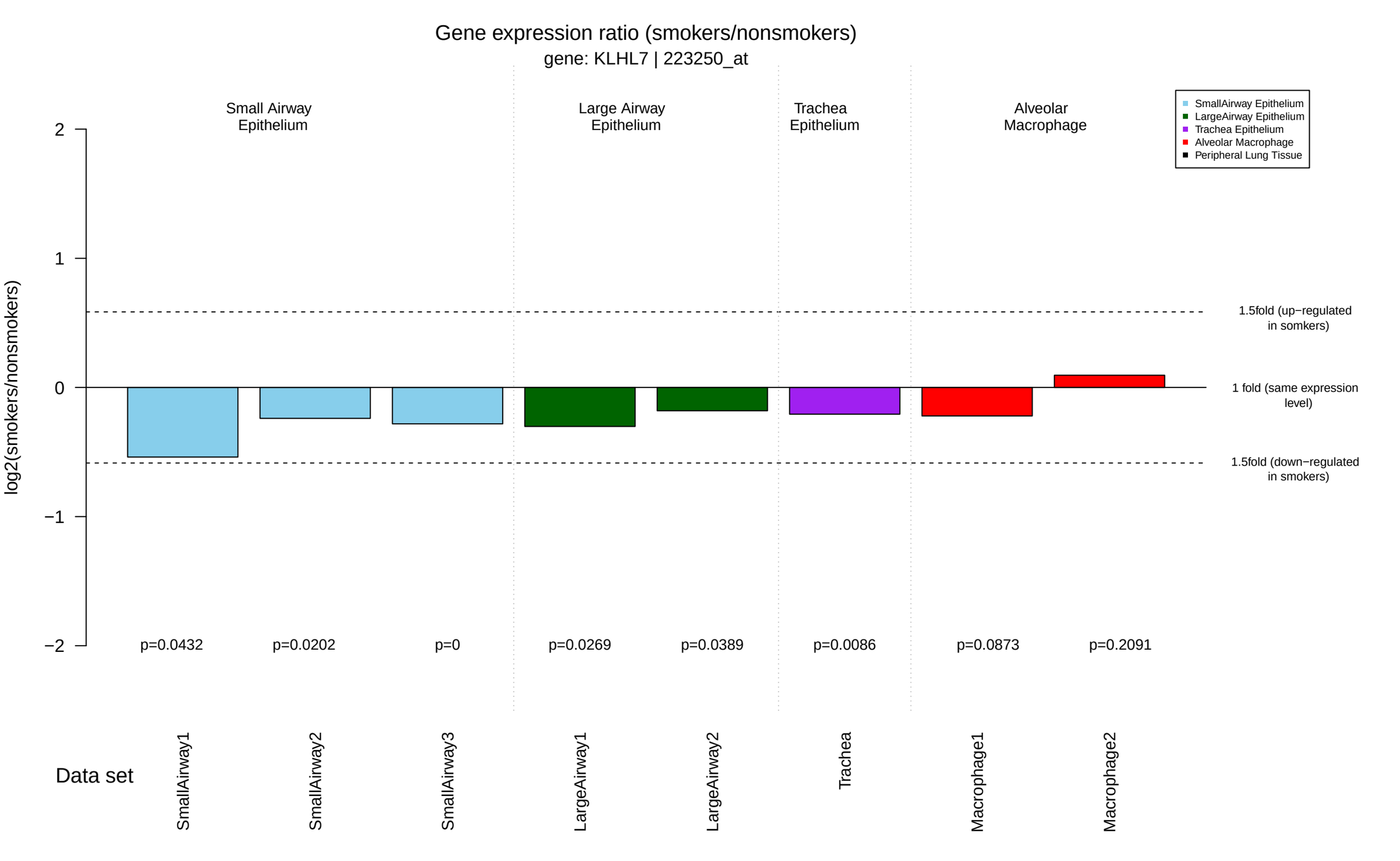
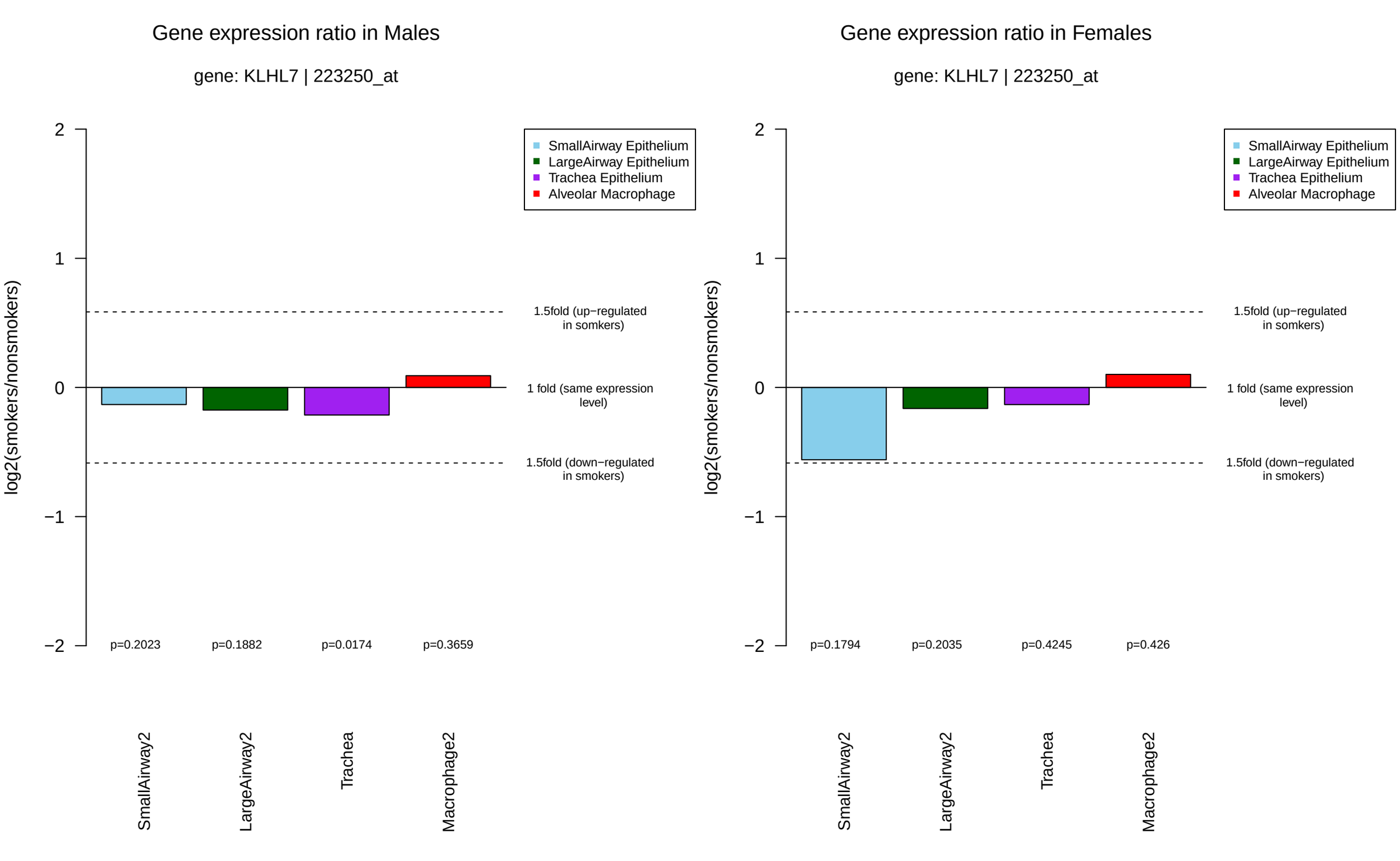
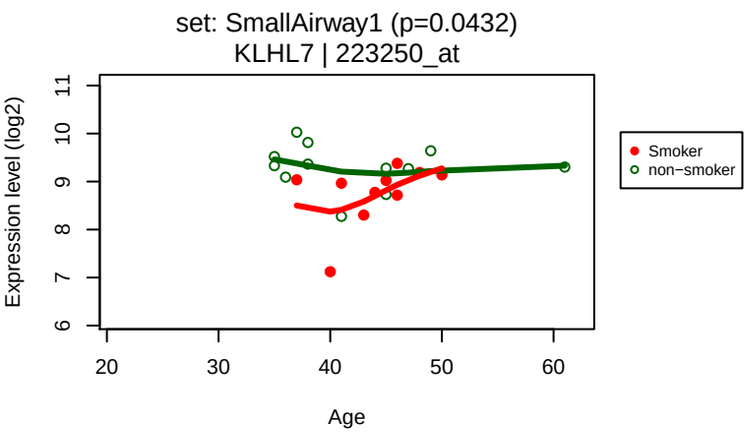
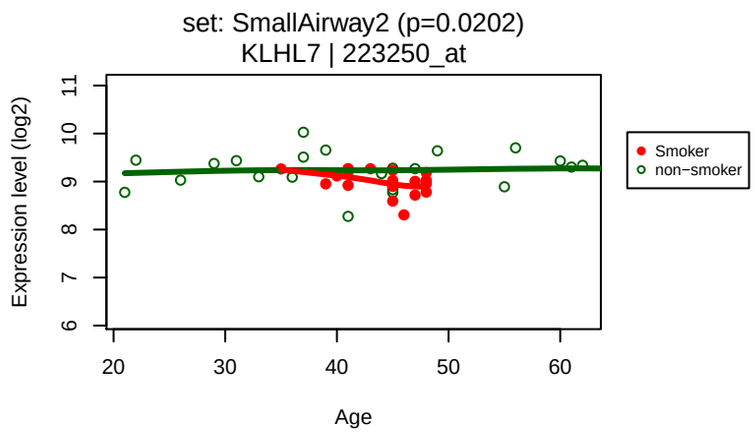
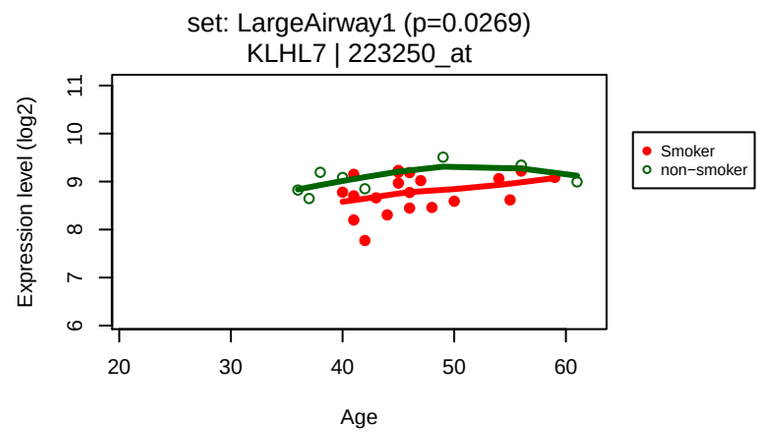

Gene expression of smokers and nonsmokers
gene: KLHL7 | 223250_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0432
p=0.0202
p=0
p=0.0269
p=0.0389
p=0.0086
p=0.0873
p=0.2091
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

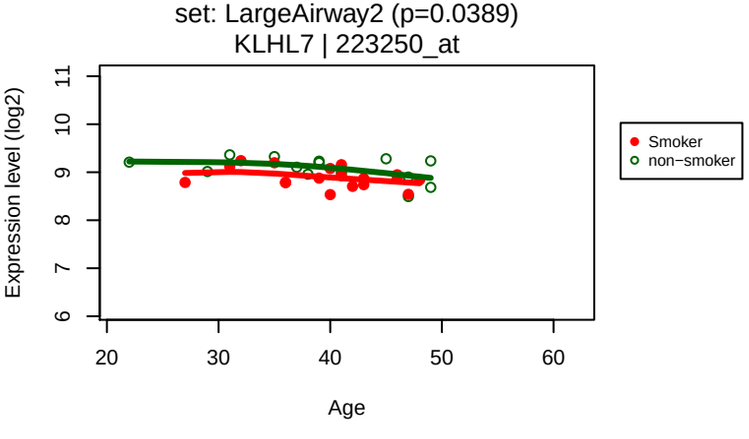

Gene expression ratio (smokers/nonsmokers)
gene: KLHL7 | 223250_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0432
p=0.0202
p=0
p=0.0269
p=0.0389
p=0.0086
p=0.0873
p=0.2091
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

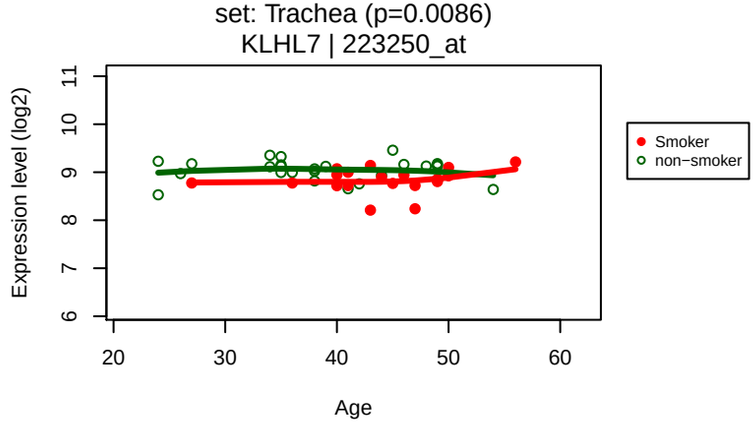

Gene expression ratio in Males
gene: KLHL7 | 223250_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2023
p=0.1882
p=0.0174
p=0.3659
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHL7 | 223250_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1794
p=0.2035
p=0.4245
p=0.426
SmallAirway2
LargeAirway2
Trachea
Macrophage2

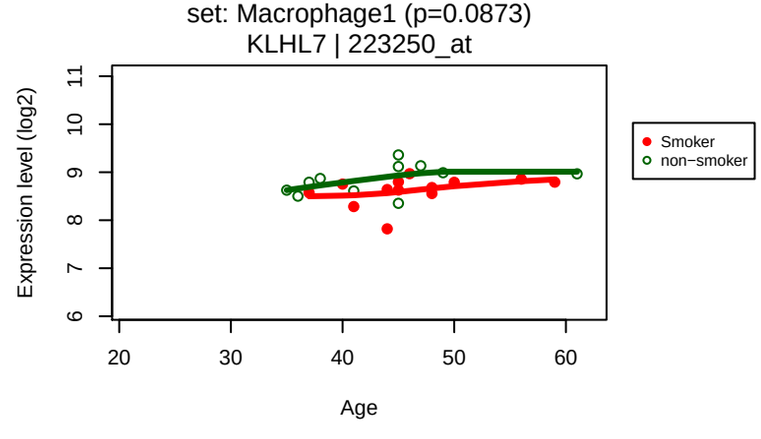

set: SmallAirway1 (p=0.0432)
KLHL7 | 223250_at
Expression level (log2)
Age
Smoker
non−smoker

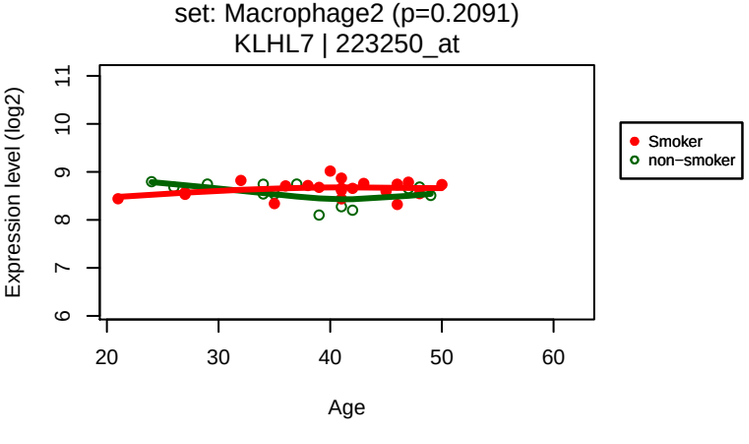

set: SmallAirway2 (p=0.0202)
KLHL7 | 223250_at
Expression level (log2)
Age
Smoker
non−smoker

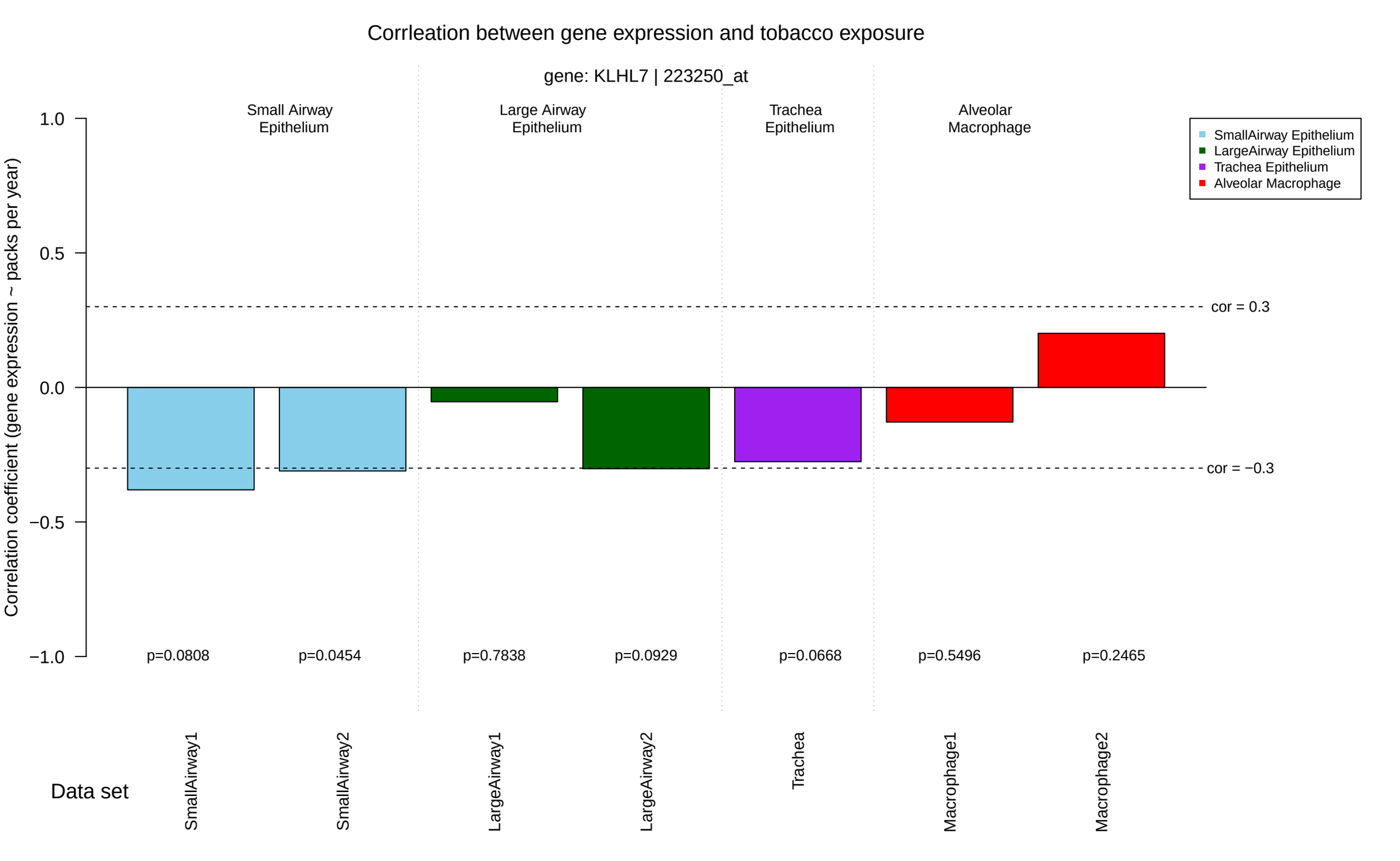

set: LargeAirway1 (p=0.0269)
KLHL7 | 223250_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0389)
KLHL7 | 223250_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0086)
KLHL7 | 223250_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0873)
KLHL7 | 223250_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2091)
KLHL7 | 223250_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHL7 | 223250_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0808
p=0.0454
p=0.7838
p=0.0929
p=0.0668
p=0.5496
p=0.2465
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2